

Configure Multi-Value Answer Routing Policy in Amazon Route 53

Summary:-

In this lab, a Multi-Value Answer Routing Policy is configured in Amazon Route 53. Multiple IP addresses are associated with the same DNS record, allowing Route 53 to return multiple healthy endpoints in response to a DNS query. The routing policy is then tested using **nslookup** to verify that multiple IP addresses are returned in a round-robin order.

Let's Start:-

- Configure a Multi-Value Answer Routing Policy.
- Create multiple DNS records with the same record name.
- Assign unique record identifiers.
- Test the routing policy using **nslookup**.
- Verify that multiple IP addresses are returned.

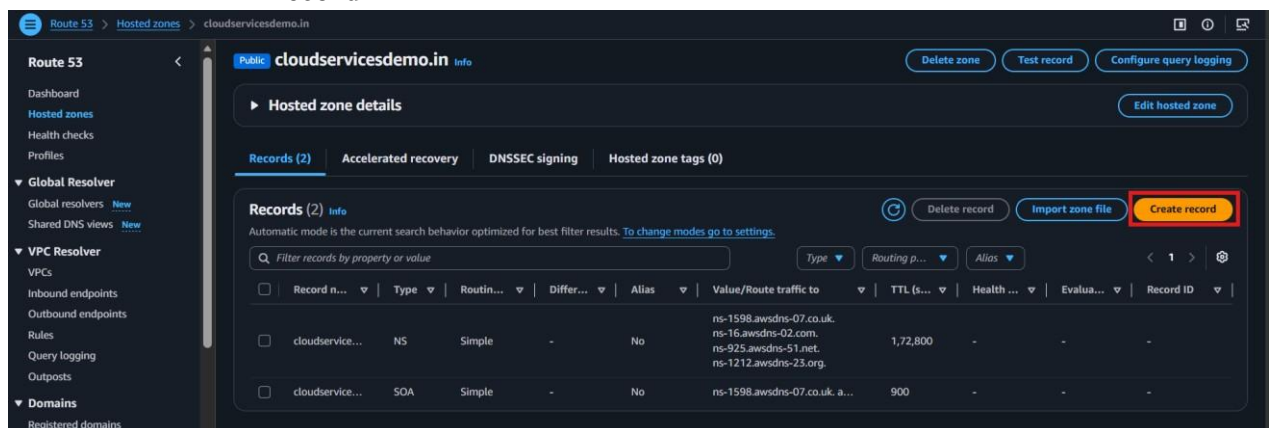
Lab Steps:-

Step 1 – Verify the Environment

- Open the **Amazon Route 53** console.
- Select an existing **Public Hosted Zone**.
- This step prepares the hosted zone required to configure the Multi-Value Answer Routing Policy.

Step 2 – Create the First Multi-Value Record

- Click **Create Record**.



- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A

- **Value:** 1.1.1.1
- **Routing Policy:** Multi-Value Answer
- **Record ID:** R1

Route 53 > Hosted zones > cloudservicesdemo.in > Create record

Quick create record Switch to wizard

▼ Record 1 Delete

Record name Info
 .cloudservicesdemo.in
Keep blank to create a record for the root domain.

☐ Alias
You can't use an alias record with multivalue routing policy.

Value Info

Enter multiple values on separate lines.

TTL (seconds) Info
 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy Info
Multivalue answer

Health check ID - optional Info
 +

Record ID Info

- Click **Create Records**.
- This step creates the first endpoint for the Multi-Value routing policy.

Step 3 – Create the Second Multi-Value Record

- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** 2.2.2.2
 - **Routing Policy:** Multi-Value Answer
 - **Record ID:** R2

Route 53 > Hosted zones > cloudservicesdemo.in > Create record

Quick create record Switch to wizard

▼ Record 1 Delete

Record name Info
 .cloudservicesdemo.in
Keep blank to create a record for the root domain.

☐ Alias
You can't use an alias record with multivalue routing policy.

Value Info

Enter multiple values on separate lines.

TTL (seconds) Info
 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy Info
Multivalue answer

Health check ID - optional Info
 +

Record ID Info

- Click **Create Records**.
- This step adds a second endpoint to the routing policy.

Step 4 – Create the Third Multi-Value Record

- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** 3.3.3.3
 - **Routing Policy:** Multi-Value Answer
 - **Record ID:** R3

The screenshot shows the AWS Route 53 'Create record' interface. The breadcrumb trail is 'Route 53 > Hosted zones > cloudservicesdemo.in > Create record'. The form fields are as follows:

- Record name:** learn (highlighted with a red box)
- Record type:** A - Routes traffic to an IPv4 address and some AWS resources (highlighted with a red box)
- Value:** 3.3.3.3 (highlighted with a red box)
- TTL (seconds):** 300
- Routing policy:** Multivalue answer (highlighted with a red box)
- Record ID:** R3 (highlighted with a red box)

At the bottom right, there is a 'Cancel' button and a highlighted orange 'Create records' button.

- Click **Create Records**.
- This step adds a third endpoint to the routing policy.

Step 5 – Verify the Multi-Value Routing Policy

- Open **Command Prompt**.
 - **Run:**
 - `nslookup learn.<your-domain-name>`
- Execute the command multiple times.

```
Windows PowerShell 5.1
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\HP> cd downloads
PS C:\Users\HP\downloads> nslookup learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:     learn.cloudservicesdemo.in
Addresses: 2.2.2.2
           1.1.1.1
           3.3.3.3

PS C:\Users\HP\downloads>
```

- Verify that all **three IP addresses** are returned.

- Observe that the order of the IP addresses changes between queries.
- This step verifies that Route 53 returns multiple IP addresses using **round-robin routing**.

Verification

- Successfully configured a Multi-Value Answer Routing Policy.
- Created multiple DNS records with the same record name.
- Assigned unique Record IDs to each endpoint.
- Verified that multiple IP addresses were returned for DNS queries.
- Confirmed round-robin ordering of the returned IP addresses.